Chess Blunder Predictor — Project Plan

Braden Hayes and Nate Poppe · Baseline Oct 28, 2026

Assignments #5 and #6. Scope follows final Designs D0–D2: completed games only (the older project description's live-scoring wording is reconciled in C9). Fall 2026 covers design deliverables; implementation runs Jan 11–Apr 23, 2027.

1. Task List (Assignment #5)

T1. Specify the Blunder-Risk Model training data, blunder label, prediction-time features, and evaluation criteria with the advisor. Owner: Braden Hayes.

T2. Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, and startup validation. Owner: Nate Poppe.

T3. Develop Game and Results Storage with database migrations, duplicate prevention, and model-versioned scores. Owner: Nate Poppe.

T4. Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels and leakage-controlled splits. Owner: Braden Hayes.

T5. Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, and bounded parallel workers. Owner: Braden Hayes.

T6. Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks and interrupted-import recovery. Owner: Nate Poppe.

T7. Develop and evaluate a baseline and candidate Blunder-Risk Model with probability-calibration checks. Owner: Braden Hayes.

T8. Develop the User Interface sign-in and import pages with progress messages and resume controls. Owner: Nate Poppe.

T9. Develop Blunder-Risk Model batch inference with feature-schema validation and versioned model loading. Owner: Braden Hayes.

T10. Develop the User Interface Games and Replay pages showing the three highest-risk moves, risk scores, and engine-best moves. Owner: Nate Poppe.

T11. Develop Risk Explanation and Analytics with plain-language risk factors and player-pattern aggregation. Owner: Braden Hayes.

T12. Test User Interface, Game Import and Validation, and Game and Results Storage end to end, including interrupted imports and user-data isolation. Owner: Nate Poppe.

T13. Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out data and the 20-game analysis target. Owner: Braden Hayes.

T14. Document and validate Configuration and User Interface deployment with a reproducible local setup, demo dataset, and user guide. Owner: Nate Poppe.

2. Milestones List (Assignment #6)

C = course deliverable, F/S = internal milestone. Canvas dates are due dates; C11–C14 are team targets for roadmap items with no published deadline. The Dec 8–Jan 22 gap is the exam and winter blackout (K2–K3); no work is planned there.

ID	Milestone	Date	Status	What it demonstrates	How verified	Verifier
C1	Professional Biography	Sep 2, 2026	Complete	Introduces each member's background and the skills each brings to the project.	Instructor checks both submitted biographies and Canvas receipts.	Instructor
C2	Team Formation and Project Description	Sep 9, 2026	Complete	Records the two-person team and the project's purpose and scope.	Roster and project-description.md compared with the submission.	Instructor
C3	Advisor Selection and Design Constraints Essay	Sep 18, 2026	Complete	Identifies the advisor and the constraints that govern the design.	Advisor confirmation and the submitted constraints essay checked.	Instructor, advisor
C4	User Stories and Use Cases	Sep 28, 2026	Complete	Defines the user workflow and the acceptance criteria later used in D2.	Submitted stories and criteria checked against D2.	Instructor
C5	Design D0: High-Level System Design	Oct 2, 2026	Complete	Defines the system modules, their interfaces, and primary owners.	D0 artifact inspected in the submission and the repository.	Instructor, advisor
C6	Design D1: Detailed Design Part 1	Oct 8, 2026	Complete	Defines storage, import, feature-extraction, and API contracts.	D1 artifact and diagrams inspected.	Instructor, advisor
C7	Design D2: Detailed Design Part 2 and Peer Review	Oct 14, 2026	Complete	Closes the design with model and UI behavior, edge cases, and peer-review responses.	D2 and its peer-review section inspected; separate notes confirmed in the repository.	Instructor, advisor
C8	Task List, Milestones, Timeline, Effort Matrix	Oct 28, 2026	Submitted	Establishes a consistent, resourced delivery baseline with owners and dates.	This PDF submitted; Tasklist.md in the repository root and plan sources in /Planning.	Instructor
C9	Requirements and Technical	Oct 30,	Planned	Freezes the completed-game scope and measurable system contracts.	Requirements traced to D0–D2; live-scoring wording resolved;	Instructor,

	Specification	2026			open ML decisions recorded.	advisor
C10	Test and Validation Plan Draft	Nov 4, 2026	Planned	Maps each requirement to fixtures, tests, metrics, and acceptance evidence.	Test cases reviewed for timing, resume, user isolation, model versions, and held-out evaluation.	Instructor
C11	Ethics review package	Nov 6, 2026	Planned	Documents fair-play boundaries, data handling, and the limits of predicted probabilities.	Advisor reviews the completed-game-only policy, data provenance, and uncertainty wording.	Advisor
F1	Modeling protocol agreed (T1)	Nov 13, 2026	Planned	Settles the data source, blunder label, pre-move feature boundary, and evaluation protocol.	Advisor reviews the T1 decision log, sample rows, and model-selection criteria.	Advisor, Nate
C12	Communication package	Nov 13, 2026	Planned	Makes the project understandable through a short narrative and draft slides.	Nate checks the usability narrative, Braden checks modeling claims, advisor reviews.	Advisor
C13	Oral presentation rehearsal 1	Nov 18, 2026	Planned	Shows a coherent explanation of the design and delivery plan.	Timed rehearsal; feedback and assigned corrections recorded.	Advisor
C14	Oral presentation rehearsal 2	Nov 25, 2026	Planned	Shows that rehearsal feedback has been incorporated.	Repeat timed presentation; major feedback items closed.	Instructor, advisor
C15	Week 14 fall report	Nov 25, 2026	Planned	Preserves the design baseline and reproduces the current planning tables.	All six components present, task IDs identical across tables, status updated.	Instructor, advisor
C16	Project Advisor Assessments	Dec 8, 2026	Planned	Gives the advisor evidence of fall design progress and readiness.	Evidence package sent Dec 4; advisor completes the assessment by Dec 8.	Advisor
S1	Startup and engine feasibility	Jan 22, 2027	Planned	The app starts with authentication, configuration, and logging working, and fixed engine settings produce features on sample PGNs.	T2/T5 smoke tests pass; an early 20-game timing probe is recorded with hardware and engine settings.	Both members
S2	Storage and training-data baseline	Feb 5, 2027	Planned	The database schema and a bounded training dataset support repeatable development.	Migrations apply cleanly on a fresh database; label samples, missing data, and split leakage audited.	Both members, advisor
S3	Resumable import and candidate model	Feb 19, 2027	Planned	Validated imports survive interruption, and a candidate model improves on the agreed baseline.	Injected import interruption leaves no duplicates; validation metrics and calibration compared with the baseline.	Both members, advisor
S4	Import-to-score pipeline	Mar 5, 2027	Planned	The import UI drives versioned inference and shows correct progress end to end.	A completed-game import runs through to stored scores; incompatible feature batches are rejected.	Both members
S5	Pre-break baseline frozen	Mar 12, 2027	Planned	A tagged working build can be resumed after midterms and spring break.	Smoke tests pass, database and model artifacts backed up, outstanding defects documented.	Both members
S6	Replay and explanations	Apr 2, 2027	Planned	Replay shows the selected board, the top-risk moves, and readable risk factors.	Acceptance fixtures show exact FENs and at least two supported factors of 25 words or fewer.	Both members, advisor
S7	Acceptance candidate	Apr 16, 2027	Planned	Core requirements pass integrated, statistical, and performance checks.	Acceptance suite and held-out evaluation pass; 20 games analyzed in under 180 s; release blockers closed.	Both members, advisor
S8	Final release and handoff	Apr 23, 2027	Planned	A repeatable local release, user guide, evaluation report, and demo are ready before finals.	Clean-environment install, demo rehearsal, and artifact hashes and documentation verified.	Advisor, both members

3. Table 1: Timeline (Assignment #6)

Each task carries 20% buffer: work is planned to finish on the "Work done" date, and the days up to "Buffered finish" are reserve. A protected block before each milestone (below) is reserved for fixes and verification only. Dependencies are finish-to-start.

ID	Task statement	Owner	Start	Work done	Buffered finish	Depends on	Path / risks
T1	Specify the Blunder-Risk Model training data, blunder label, prediction-time features, and evaluation criteria with the advisor.	Braden Hayes	Oct 29, 2026	Nov 6, 2026	Nov 10, 2026	C7	Model; R1, R2

T2	Develop Authentication, Configuration, and Logging with Lichess sign-in, session checks, and startup validation.	Nate Poppe	Jan 11, 2027	Jan 15, 2027	Jan 20, 2027	C7	App; R3, R7
T3	Develop Game and Results Storage with database migrations, duplicate prevention, and model-versioned scores.	Nate Poppe	Jan 25, 2027	Feb 1, 2027	Feb 3, 2027	T2	App; R7
T4	Obtain and prepare the Blunder-Risk Model training dataset with reproducible labels and leakage-controlled splits.	Braden Hayes	Jan 25, 2027	Feb 1, 2027	Feb 3, 2027	T1, T5	Model; R1, R2
T5	Develop Chess Feature Extraction with Stockfish analysis, reusable engine results, and bounded parallel workers.	Braden Hayes	Jan 11, 2027	Jan 15, 2027	Jan 20, 2027	T1	Model; R4
T6	Develop Game Import and Validation for Lichess and PGN inputs with completed-game checks and interrupted-import recovery.	Nate Poppe	Feb 8, 2027	Feb 15, 2027	Feb 17, 2027	T2, T3	App; R3, R7
T7	Develop and evaluate a baseline and candidate Blunder-Risk Model with probability-calibration checks.	Braden Hayes	Feb 8, 2027	Feb 15, 2027	Feb 17, 2027	T4	Model; R1, R2
T8	Develop the User Interface sign-in and import pages with progress messages and resume controls.	Nate Poppe	Feb 22, 2027	Mar 1, 2027	Mar 3, 2027	T2, T6	App; R3
T9	Develop Blunder-Risk Model batch inference with feature-schema validation and versioned model loading.	Braden Hayes	Feb 22, 2027	Mar 1, 2027	Mar 3, 2027	T3, T5, T7	Model; R2, R7
T10	Develop the User Interface Games and Replay pages showing the three highest-risk moves, risk scores, and engine-best moves.	Nate Poppe	Mar 22, 2027	Mar 29, 2027	Mar 31, 2027	T3, T8, T9	App; R5, R7
T11	Develop Risk Explanation and Analytics with plain-language risk factors and player-pattern aggregation.	Braden Hayes	Mar 22, 2027	Mar 29, 2027	Mar 31, 2027	T9	Model; R2, R5
T12	Test User Interface, Game Import and Validation, and Game and Results Storage end to end, including interrupted imports and user-data isolation.	Nate Poppe	Apr 5, 2027	Apr 12, 2027	Apr 14, 2027	T6, T8, T10, T11	App; R3, R7
T13	Validate and refine Chess Feature Extraction, Blunder-Risk Model, and Risk Explanation and Analytics against held-out data and the 20-game analysis target.	Braden Hayes	Apr 5, 2027	Apr 12, 2027	Apr 14, 2027	T5, T7, T9–T11	Model; R2, R4
T14	Document and validate Configuration and User Interface deployment with a reproducible local setup, demo dataset, and user guide.	Nate Poppe	Apr 19, 2027	Apr 20, 2027	Apr 21, 2027	T12, T13, S7	Release; R6–R8

Critical paths. Model: T1 → T5 → T4 → T7 → T9 → T11 → T13 → T14 → S8. App: T2 → T3 → T6 → T8 → T10 → T12 → T14 → S8. Both are critical because each member carries one path at full capacity; a slip on either consumes a protected block or cuts optional scope (T11 player patterns go first).

Milestone dates and protected blocks

Milestone	Planned completion	Protected block
C1–C7	Sep 2–Oct 14, 2026	Complete
C8	Oct 28, 2026	Oct 28
C9	Oct 30, 2026	Oct 29
C10	Nov 4, 2026	Nov 2–3
C11	Nov 6, 2026	Nov 5
C12, F1	Nov 13, 2026	Nov 12
C13	Nov 18, 2026	Nov 16–17
C14, C15	Nov 25, 2026	Nov 23–24
C16	Dec 8, 2026	Dec 3–4
S1	Jan 22, 2027	Jan 21
S2	Feb 5, 2027	Feb 4
S3	Feb 19, 2027	Feb 18
S4	Mar 5, 2027	Mar 4
S5	Mar 12, 2027	Mar 11
S6	Apr 2, 2027	Apr 1
S7	Apr 16, 2027	Apr 15

S8	Apr 23, 2027	Apr 22

Availability constraints

ID	Dates	Constraint	Response
K1	Nov 26–29, 2026	Thanksgiving	No project work
K2	Dec 6–12, 2026	Fall exams	No development; C16 evidence sent Dec 4
K3	Dec 13, 2026–Jan 10, 2027	Winter break	No development; restart Jan 11
K4	Jan 18, 2027	University holiday	T2/T5 fit the remaining days
K5	Mar 8–12, 2027	Midterms	6 h/member; freeze and repair only
K6	Mar 15–21, 2027	Spring break	No project work
K7	Apr 24–29, 2027	Spring finals	No development; release Apr 23
K8	Jan 11–Apr 23, 2027	Course load and co-op	Braden 18 credits; Nate 15 credits plus 20 h/week co-op; 12 project h/week each

4. Table 2: Effort Matrix (Assignment #6)

DRAFT: estimates pending team review on Oct 29. Each owner estimates optimistic, most likely, and pessimistic hours (O/M/P); the matrix uses (O + 4M + P) / 6. Support hours are the other member's review and pairing time. **P** marks the primary owner, who carries the largest share of every task. Base hours are shown per task, with the 20% contingency added once at the bottom.

ID	Task	Owner O / M / P (h)	Braden Hayes (h)	Nate Poppe (h)	Row total (h)
T1	Modeling protocol	6 / 10 / 16	**10.3 P**	2.0	12.3
T2	Auth, configuration, logging	8 / 12 / 20	1.0	**12.7 P**	13.7
T3	Game and results storage	8 / 11 / 18	2.0	**11.7 P**	13.7
T4	Training dataset	8 / 13 / 22	**13.7 P**	1.0	14.7
T5	Chess feature extraction	10 / 14 / 24	**15.0 P**	1.0	16.0
T6	Game import and validation	10 / 15 / 26	2.0	**16.0 P**	18.0
T7	Baseline and candidate model	9 / 14 / 22	**14.5 P**	1.0	15.5
T8	Sign-in and import UI	7 / 11 / 18	1.0	**11.5 P**	12.5
T9	Batch inference	8 / 12 / 18	**12.3 P**	3.0	15.3
T10	Games and replay UI	9 / 14 / 22	2.0	**14.5 P**	16.5
T11	Risk explanation and analytics	8 / 12 / 20	**12.7 P**	2.0	14.7
T12	App integration testing	8 / 12 / 20	4.0	**12.7 P**	16.7
T13	Model validation and refinement	8 / 13 / 20	**13.3 P**	3.0	16.3
T14	Deployment and user guide	4 / 7 / 12	1.5	**7.3 P**	8.8
Base subtotal			**105.3**	**99.4**	**204.7**
Contingency (+20%)			21.1	19.9	41.0
Total			**126.4**	**119.3**	**245.7**

Braden carries 5.9 more base hours (about 6%), driven by the model path. To rebalance, Nate takes up to 3 hours of T13 test execution; if Nate's availability drops instead, T12 fixture work moves to Braden. Primary ownership does not change.

5. Work Breakdown Structure

1. **Scope and foundations**

- T1. Modeling protocol (Braden Hayes)
- T2. Auth, configuration, logging (Nate Poppe)
- T3. Game and results storage (Nate Poppe)

2. **Data and model development**
   - T4. Training dataset (Braden Hayes)
   - T5. Chess feature extraction (Braden Hayes)
   - T7. Baseline and candidate model (Braden Hayes)
3. **Import, scoring, and user experience**
   - T6. Game import and validation (Nate Poppe)
   - T8. Sign-in and import UI (Nate Poppe)
   - T9. Batch inference (Braden Hayes)
   - T10. Games and replay UI (Nate Poppe)
   - T11. Risk explanation and analytics (Braden Hayes)
4. **Validation and delivery**
   - T12. App integration testing (Nate Poppe)
   - T13. Model validation and refinement (Braden Hayes)
   - T14. Deployment and user guide (Nate Poppe)

6. Risk Register

Likelihood: low <25%, medium 25–60%, high >60%. High impact threatens a core requirement or milestone. Owners check triggers weekly; the last column shows where each risk appears in the schedule.

ID	Risk	Category	Likelihood	Impact	Strategy	Mitigation	Trigger	Owner	In Table 1
R1	Training data, labels, or advisor decision arrive late	Technical / external	Medium	High	Reduce	Fix label and protocol in T1; start with a small reproducible sample and a simple baseline	No labeled sample or advisor decision by F1 (Nov 13)	Braden	T1, T4, T7; F1, S2
R2	Leakage or weak calibration invalidates predictions	Technical	Medium	High	Reduce	Keep played-move outcomes out of features; split by game and time; audit calibration	Leakage found, or candidate fails criteria at S3	Braden	T4, T7, T9, T13
R3	Lichess sign-in, rate limits, or resume logic block import	External	Medium	High	Reduce	Verify API early; test interruption with fixtures; keep PGN upload as demo fallback	Sign-in fails at S1, or resume duplicates games at S3	Nate	T2, T6, T8, T12
R4	Stockfish analysis exceeds 180 s for 20 games	Technical	High	High	Reduce	Time early in T5; cache results, cap workers, tune search depth	20-game probe projects over 180 s at S1	Braden	T5, T13; S1, S7
R5	Optional features or model experiments displace core work	Team / scope	Medium	Medium	Avoid	One baseline and one candidate only; defer player patterns if core slips	A core task uses all its buffer, or replay incomplete by Mar 29	Braden	T7, T10, T11; S6
R6	Course load, co-op, exams, or breaks reduce availability	Team	High	High	Reduce	12 h/week caps, weekly availability check, shift support hours before using milestone blocks	Either member under 8 available hours for two straight weeks	Nate	K1–K8, all buffers
R7	Integration defects or environment differences delay release	Technical / team	Medium	High	Reduce	Shared API and feature fixtures before parallel work; clean-install test	Contract tests fail at S4/S6, or clean install fails before S8	Nate	T3, T9–T14
R8	Course deadlines are published earlier than team targets	External	Medium	High	Reduce	Check Canvas weekly; pull work earlier if needed	A new deadline precedes a team target	Nate	C11–C15, S8